

HW3 – report

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1. Output for 'python3 NaiveBayes.py'

RuntimeWarning: divide by zero encountered in log

Total Sentences correctly: 775

Predicted correctly: 580

Accuracy: 74.83871%

2. Output for 'python3 NaiveBayes_improved.py'

Total Sentences correctly: 775

Predicted correctly: 687

Accuracy: 88.64516%

3. Improved version

In the improved version, I added Laplace smoothing to avoid zero probabilities. This method ensures that every word receives a small positive likelihood of appearing and a small likelihood of not appearing, even if the word never appears or always appears in a particular class in the training data. In the program, when I calculate the probability of a word in a class, I add 1 to the word count and add 2 to the denominator, which use the rule $\frac{(\text{count} + 1)}{(N_c + 2)}$. This prevents the model from getting a probability of exactly 0 or 1, which used to cause errors when taking the log. I also added a very small constant during the log calculation to prevent the model from taking $\log(0)$. After these changes, the warnings removed and the accuracy increased.